Network and System Defence
Final Projects AY 2024/2025

General rules for all projects:

Addressing:

- Assign the IP addresses at will, but pay attention to the IP address pool you choose:
    - Private address ranges for private networks
    - Public address ranges for the ASes
    - /30 private networks for point-to-point links between routers

Legend:

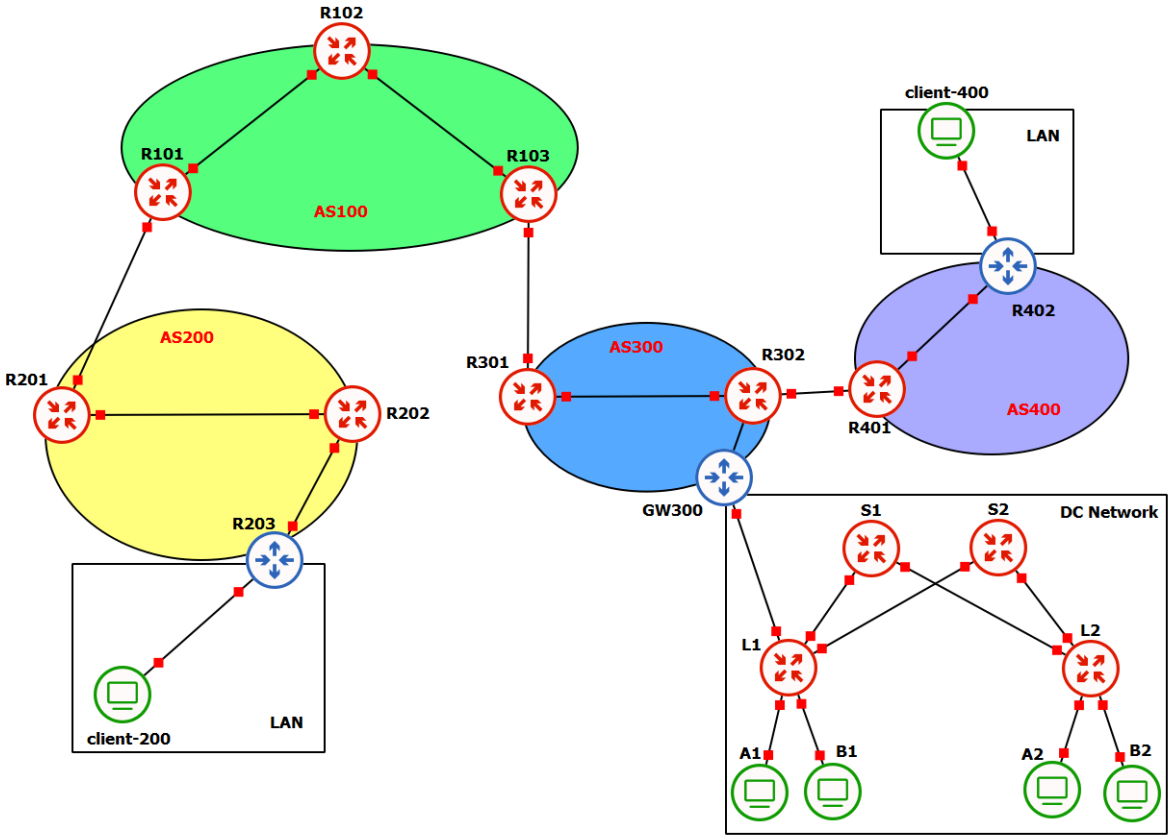
- Red nodes are FRR routers (nsdcourse/frr:latest)
- Green nodes are simple clients (nsdcourse/basenet:latest)
- Blue nodes are Linux routers (nsdcourse/basenet:latest)
- Blue switches (with the 3D icon) are simple GNS3 L2 switches
- eBPF switches (with the bee icon 🐝) are eBPF enabled images (nsdcourse/eBPF:latest)

Delivery:

- The project must be deliverable respecting the deadlines provided in the Teams Channel
- What should be delivered:
    - A presentation of the project where you explain the work done
        - Formatted as a Readme, slides, or a pdf document
    - The relevant configurations formatted either as:
        - A dump of the nodes configurations/programs/scripts in text file
        - or an exported gns3 project with persistent configurations of the nodes
            - Note: do not include the docker images

NOTE: Any assignment not explicitly specified in this document is up to your discretion.

Project #1



AS100

- AS100 is a transit Autonomous System providing network access to two customers: AS200 and AS300
    - Configure eBGP peering with AS200 and AS300
    - Configure iBGP peering between border routers
    - Configure OSPF
    - Configure LDP/MPLS in the core network

AS200

AS 200 is a customer AS connected to AS100, which provides transit services.

- Setup eBGP peering with AS100
- Configure iBGP peering
- Configure internal routing as you wish (with or without OSPF)
- ***R203*** is **not** a BGP speaker
    - It has a default route towards R202
    - It has a public IP address from the IP address pool of AS200
    - It is the Access Gateway for the LAN attached to it
        - Configure dynamic NAT

        - Configure a simple firewall to allow just connections initiated from the LAN

Client 200

- This device is sensitive and must be configured to use Falco for runtime security monitoring. Show at least **two** meaningful policies used to detect security related events. Possible policies can include:
    - Detecting attempts to access sensitive files or directories not normally accessed by the client.
    - Alerting on unexpected privilege escalation attempts, such as processes trying to run with elevated privileges outside their normal behavior.
    - Monitoring for abnormal outbound network connections, for instance, processes connecting to IP addresses not previously seen or deemed suspicious.
- OpenVPN → see later dedicated section.

AS300

AS 300 is a customer AS connected to AS 100, which provides transit services. It also has a lateral peering relationship with AS 400.

- Setup eBGP peering with AS400 and AS100
- Configure iBGP peering
- Configure internal routing as you wish (with or without OSPF)
- ***GW300*** is **not** a BGP speaker
    - It has a default route towards R302
    - It has a public IP address from the IP address pool of AS300
    - It is the Access Gateway for the Data Center network attached to it
        - Configure dynamic NAT
        - OpenVPN server → see later dedicated section

DC Network

DC Network is a leaf-spine Data Center network with two leaves and two spines. There are 2 tenants (A and B) in the cloud network, each hosting two virtual machines connected to leaf1 and leaf2. The tenants are assigned **one** broadcast domain each.

- Realize VXLAN/EVPN forwarding in the DC network to provide L2VPNs between the tenants' machines
- In L1, enable the connectivity to the external network:
    - Tenant A must reach the internet through (hint: L3VNI, EVPN Type5 route for the default net)
    - Tenant B machines must be reachable through OpenVPN, but they cannot access the internet directly (unlike Tenant A)

AS400

AS 400 has a lateral peering relationship with AS 300.

- Setup eBGP peering with AS400 and AS100
- ***R402*** is **not** a BGP speaker
    - It has a default route towards R401
    - It has a public IP address from the IP address pool of AS400
    - It is the Access Gateway for the LAN attached to it
        - Configure dynamic NAT
        - Configure OpenVPN→ see later dedicated section

Client 400

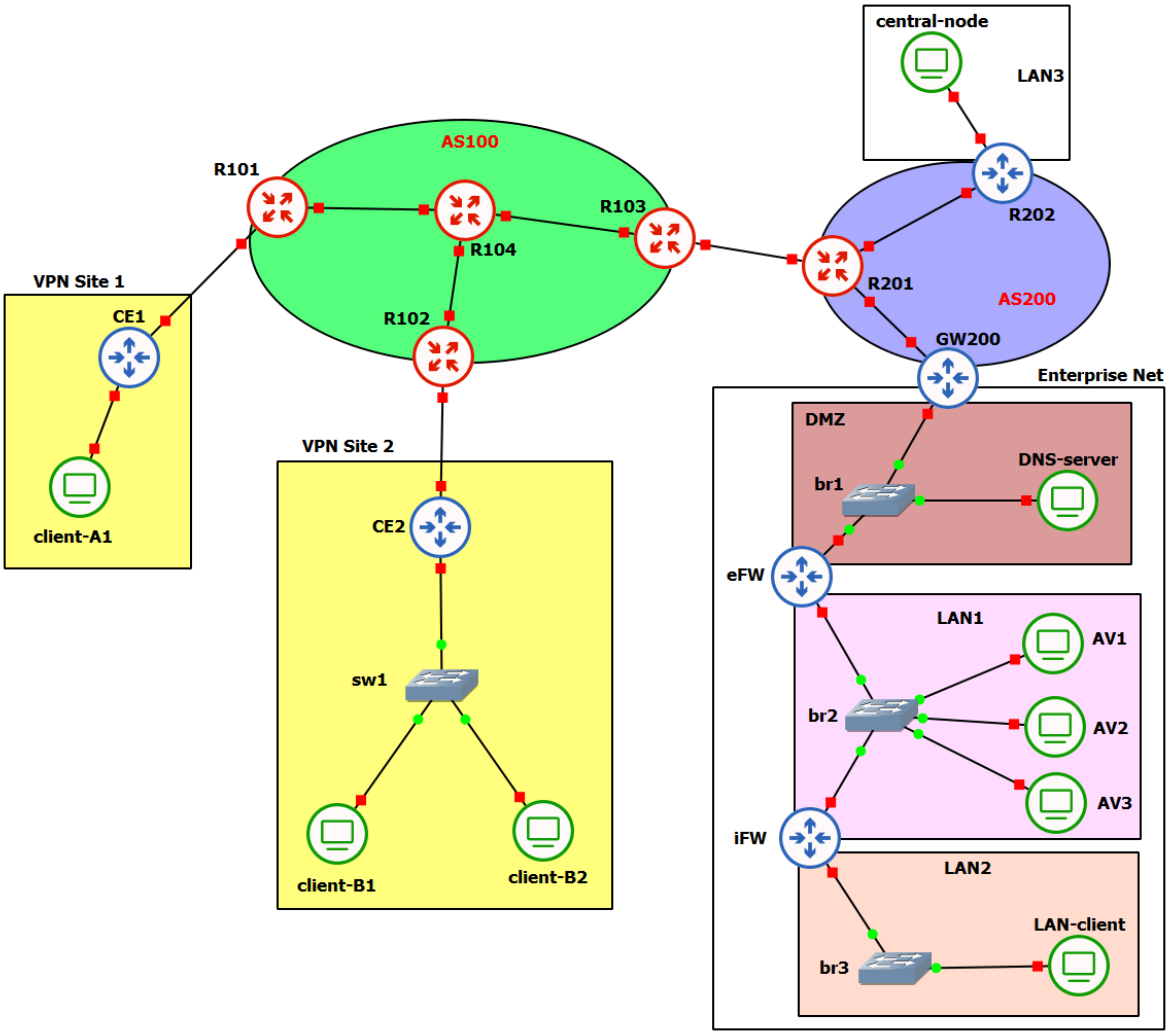
- This is a simple LAN device with a default route through R402.

OPENVPN

Setup OpenVPN to realize a VPN between client-200, R402's LAN, and the DataCenter network.

- **Client-200** is an OpenVPN client
- **R402** is an OpenVPN client, providing VPN access to and from the LAN attached to it.
- **GW300** is the OpenVPN server, providing VPN access to and from the Data Center network. In particular, only the network belonging to tenant A must be accessible through the VPN.
    - Other Clients (Client-200 and R402) must receive a route from the server to reach the datacenter VM network.

Project #2



AS 100

- AS100 has a BGP peering relationship with AS200 and provides a BGP/MPLS VPN to two sites of one customer (in yellow).
    - Configure eBGP peering with AS200
    - Configure iBGP peering between border routers
    - Configure OSPF
    - Configure LDP/MPLS in the core network
    - R101 is the provider edge for VPN Site 1
    - R102 is the provider edge for VPN Site 2

VPN Site 1

VPN Site 1 has just one customer edge (CE1) and one LAN client (client-A1).

VPN Site 2

VPN Site 2 has one customer edge (CE2) and two clients in the LAN (client-B1 and client-B2).
Configure MACSEC in the LAN with MKA.

AS200

AS 200 has a lateral peering relationship with AS 100.

- Setup eBGP peering with AS100
- ***R202*** is **not** a BGP speaker
    - It has a default route towards R201
    - It implements a VPN with IPSEC towards eFW → see later dedicated section
- ***GW200*** is **not** a BGP speaker
    - It has a default route towards R201
    - It is the edge gateway for the Enterprise Network behind it

LAN3

LAN3 is a simple LAN with one device (the central-node).

Enterprise Network

In the Enterprise network, we have multiple sub-networks interconnected by some L3 devices:

- **DMZ** is the Demilitarized Zone
    - It is a sub-network with address range taken from the AS200 IP pool
    - It hosts a **DNS server** which implements the following services:
        - DNSSEC enabled authoritative nameserver for "***nsdcourse.xyz***", which publishes the records for the "***www***" subdomain
            - NOTE: Obviously, you don't ***actually*** register the domain. Consequently, you do not have to publish any DS record in any parent zone
        - Apache2 simple web server accessible through "www.nsdcourse.xyz"
    - **eFW** is a Linux router implementing an external Firewall and forwarding packets between the DMZ and LAN1
- **LAN1** is the LAN which hosts three antivirus runners → see later dedicated section
    - **iFW** is a Linux router implementing an internal Firewall and forwarding packets between the LAN1 and LAN2
- **LAN2** is a simple LAN with one client device (LAN-client)

- **Security Policy for the Firewalls. Configure the Firewalls and the GW200 accordingly:**
    - LAN-client can access the external network (both the "Internet" and the DNS server) only the connection is originated by it
    - Traffic from/to the AVs is enabled exclusively between the AVs and the *central-node* in LAN3
    - From the external network, allow inbound connections for the following types of traffic:
        - DNS requests to *DNS-server*

- HTTP traffic to *DNS-server*
- IPSEC traffic to *eFW*

IPsec configuration

Configure an IPSec tunnel between R202 and eFW to create a site-to-site VPN between LAN3 and LAN1, using strongswan. This is needed for the end-to-end communication between the central node and the AVs.

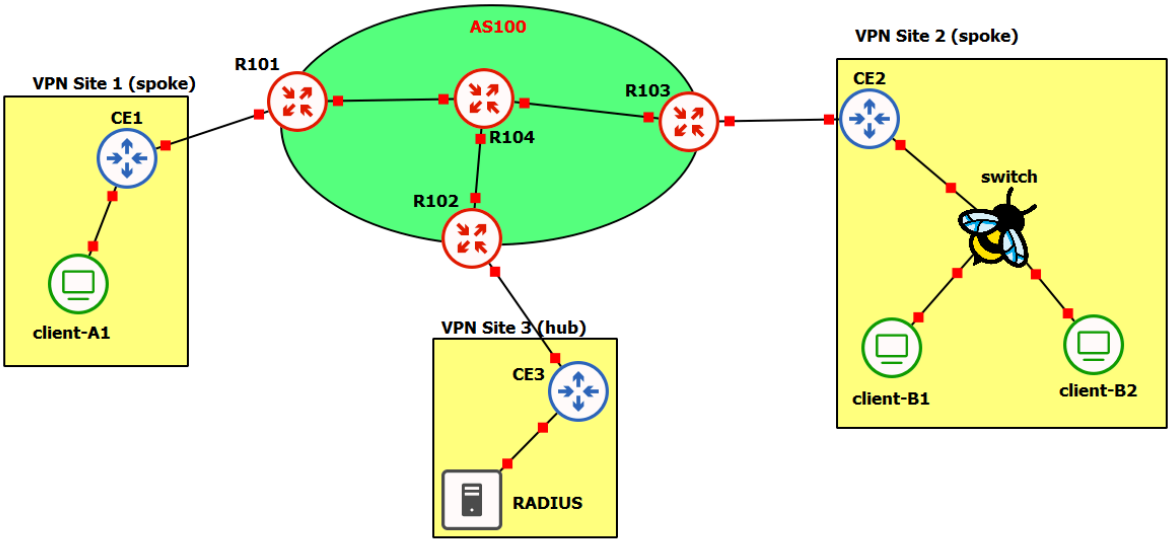
Antivirus Runners and Central Node

You are expected to create a virtualized environment to test binaries for the presence of malware. In particular, you have to setup three virtual machines or containers in LAN1, each hosting a different AV of your choice.
A central node (in LAN3) will allow dropping an executable and distributing it to the various testing nodes. The testing nodes will scan/run the executables and deliver results to the central node, which will build a report to the user, showing what threats (if any) were discovered in the binary.
Given the criticality of this infrastructure, certain precautions must be taken:

1. Runner nodes must be subject to snapshots so that they can be restarted in a 'clean' state each time a new scan has to be started.
2. To prevent the exfiltration of threats on the network, nodes must be protected by firewalls. Refer to the “Enterprise Networks” section for the specific policies.

In the central node, if needed, setup an external internet connection through a NAT GNS3 node (not displayed in the figure).

Project #3



AS 100

- AS100 provides a BGP/MPLS VPN to three sites of one customer (in yellow).
    - Configure iBGP peering between border routers
    - Configure OSPF
    - Configure LDP/MPLS in the core network
    - Configure a BGP/MPLS VPN with an hub and spoke topology:
        - R101 is the provider edge for VPN Site 1 (spoke)
        - R102 is the provider edge for VPN Site 3 (hub)
        - R103 is the provider edge for VPN Site 2 (spoke)
        - spoke-to-spoke communication is enabled through the hub
        - CE-PE route advertisement is realized with *automatic routing protocols* (BGP or OSPF)

VPN Site 1

VPN Site 1 has just one customer edge (CE1) and one LAN client (client-A1).
Client-A1 device is sensitive, so it must be configured to use ***Mandatory Access Control***.

VPN Site 3

VPN Site 3 has one customer edge (CE2) and one Radius server, configured to serve Access Requests from Site 3 supplicants (client-B1, client-B2).

VPN Site 2

VPN Site 2 has one customer edge (CE3) connecting a LAN to the BGP/MPLS VPN.

The switch is the authenticator of that LAN serving two clients (B1 and B2), communicating with the Radius server in Site 3.

- Configure 802.1x authentication with hostapd in the authenticator and wpa_supplicant in the clients.
- In the switch, deploy an *XDP program* and its associated *userspace* program which parses the Radius/802.1x messages to extract the Radius authentication information (i.e. Authentication Success messages). With this information, configure the Linux bridge according to the outcome of the authentication messages: grant/revoke access to a station based on MAC address (with ebtables).
    - The Radius server is configured to enforce VLAN assignment to the stations:
        - Client-B1 will get vlan id 32
        - Client-B2 will get vlan id 95
    - In the *XDP program*, parse the incoming Radius messages to extract the information about the assigned VLAN. This information is then stored in a map and used by the *userspace* program to assign the VLAN to the clients when a successful authentication packet arrives.
        - Note: you can use whatever language for the *userspace* application
        - Note: the *userspace* program can cooperate with hostapd events, or the events can be handled entirely by the custom *userspace* program.
        - Note: there is no need to make a *complete* implementation handling all the possible events defined in the protocol. It is suggested to focus mainly on the events required to realize the above described features.